Regime-Adaptive Multimodal Transformer for Cross-Market Equity Forecasting

Authors
Shivansh Gupta (230054)
Vivek (230119)

Phase & Methods
Phase 3 — Chronos-T5 + LoRA + HMM

Institution
Rishihood University

Agenda

Overview

Where Phase 2 left off

RAMT collapsed on OOS
A foundation model fixes the root cause

Vanishing margins

- When σ(predictions) → 0, pairwise margins → 0 and gradients carry no cross-sectional information.
- The MSE anchor pulled predictions further toward the mean instead of regularizing them.

Capacity vs. data

- A 97k-parameter transformer learning a 200-stock cross-section over only 26 monthly rebalances per fold is over-parameterized for the event volume.
- Pretrained foundation models bypass this.

Black-box fragility

- Even when gradients flowed, attention maps were diffuse and uninterpretable.
- A frozen pretrained backbone with thin adapters surfaces the inductive bias we lacked.

Literature review

03

Literature review

Time-series foundation models

- Ansari et al. 2024 (Chronos): tokenize numerical sequences and adapt T5 transformers
- Pretrained on synthetic + real time-series corpora
- Zero-shot performance often matches task-specific models

Parameter-efficient fine-tuning

- Hu et al. 2021 (LoRA): inject low-rank decomposition $W = W_0 + BA$
- Applied to attention projection layers
- <1% trainable parameters, no catastrophic forgetting

Hybrid ML/DL systems for finance

- López de Prado (2018): meta-labelling and ensemble methods
- Guidolin & Timmermann (2007): regime-switching for portfolio allocation
- Motivates the HMM gate in our hybrid architecture

Phase 3 research timeline

Six iterations from foundation-model adoption to the final hybrid

01

Chronos-T5 zero-shot

Backbone evaluated as-is on monthly alpha. Directional accuracy ~50%, no edge.

02

LoRA adapters (r=8)

Inject low-rank adapters into Chronos attention layers. ~0.1% trainable params. IC turns positive.

03

Foundation-Only backtest

Chronos-LoRA used as sole alpha signal, monthly rebalance. Sharpe 1.34, CAGR 23.5%.

04

Simple Hybrid (50/50)

Equal-weight blend of momentum + Chronos. Sharpe 0.91. Lower but smoother.

05

Triple-Expert + HMM gate

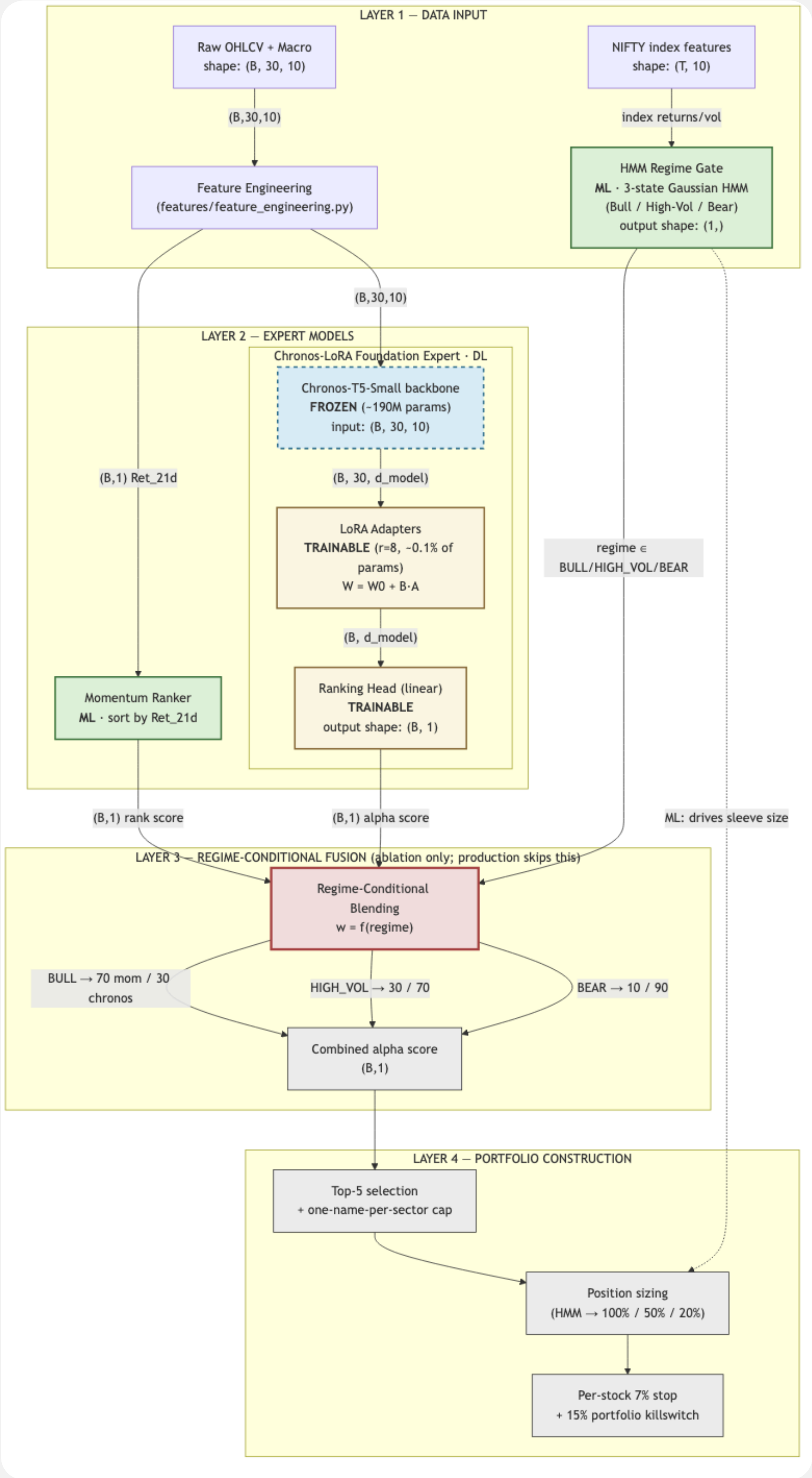
Adds HMM regime sizing. Sharpe drops to 0.54 in the 2024-26 bull. Diagnostic signal.

06

4-window HMM ablation

HMM tested across 2008-10, 2010-12, 2013-15, 2024-26. Acts as conditional insurance.

Hybrid architecture



Frozen foundation backbone + thin LoRA + classical signals + HMM regime gate.

Backtest results

Headline result — Triple-Expert hybrid

NIFTY 200, 2024-01-10 to 2026-01-27, 25 monthly rebalances, 22 bps friction

Sharpe (net)	CAGR	Max drawdown	Win rate
0.54	+8.2%	−13.8%	57.7%

Component ablation

Same OOS window, rebalance dates, friction, and stops.
'Hybrid –' rows isolate each component's contribution.

Scenario	Sharpe	CAGR	Max DD
Momentum + HMM (production)	0.83	13.5%	-18.7%
Foundation Only (Chronos-LoRA)	1.34	23.5%	-16.0%
Simple Hybrid (Mom + Chronos, 50/50)	0.91	22.8%	-11.1%
Triple-Expert (Mom + Chronos + HMM)	0.54	8.2%	-13.8%
Hybrid – Momentum (Chronos + HMM only)	0.68	13.5%	-13.7%
Win rate (all scenarios)			64.0% | 61.5% | 57.7% | 57.7% | 50.0%

HMM regime gate — across-regime study

Same momentum signal, with vs without HMM regime sizing, four out-of-sample windows.

Window	HMM Sharpe	Flat Sharpe	HMM Max DD	Flat Max DD
2008–2010	0.25	0.17	-43.2%	-52.7%
2010–2012	0.79	-3.00	-12.2%	-30.1%
2013–2015	0.84	0.59	-33.2%	-33.2%
2024–2026	0.66	1.35	-18.7%	-19.7%

HMM is conditional insurance — it pays for itself in stressed regimes (2008–10 saves 9.4 pp drawdown; 2010–12 turns a negative Sharpe into +0.79) and costs upside in clean bulls (2024–26).

What worked

Three concrete components that each carry their own ablation row. None of these required ML inference at deploy time except step 02.

01	Foundation backbone	Frozen Chronos-T5-Small (190M params). Zero-shot directional accuracy near 50%; LoRA lifts to 47.0% with positive IC.
02	LoRA adapters (r=8)	~0.1% of weights trainable. No catastrophic forgetting. Best single-signal Sharpe (1.34) in the test window.
03	HMM regime gate	Conditional insurance. Saves drawdown in crises, caps upside in clean bulls. Verified across four OOS windows.

What worked vs what didn't

Worked

Foundation-model swap. Replacing the bespoke transformer with frozen Chronos + LoRA fixed the prediction-collapse failure.
Component-level ablation. Five scenarios on identical OOS window made each component's contribution legible.
4-window HMM study. Confirmed regime gating is conditional, not decorative.

Didn't

Naive triple-blend. Triple-Expert (Sharpe 0.54) underperformed Foundation-Only (1.34) in the 2024-26 bull because HMM caps upside.
Static fusion weights. The 50/50 simple hybrid has no learned gating; differentiable gating remains open.
Single-signal interpretability. Chronos attention maps over its tokenizer are not directly interpretable in price-units.

Reproducibility & engineering

05

Reproducibility

Turn-key review

Clone, build, and reach the dashboard in three commands. Every dependency, dataset, and pipeline is pinned and versioned.

Limitations

Window length

- One 2-year OOS window (2024–26) plus three historical windows for HMM
- Confidence intervals on Sharpe estimates are wide
- Ideally evaluate over 5+ rolling windows

Foundation model cost

- Chronos-T5-Small inference at every rebalance is tractable
- Adds latency versus classical signals
- Larger Chronos variants improve accuracy at material cost

Static fusion

- HMM gate uses hand-set weights per regime
- A learned (differentiable) gating function is the natural next step
- Would let the model adapt weights from data, not heuristics

What's next

Each item below maps to a specific failure mode surfaced in this phase.

Differentiable regime gating

Replace hand-set HMM weights with a small MLP gating head trained jointly with LoRA adapters.

Wider OOS evaluation

Walk-forward over 2015-2026 in 6-month folds; report Sharpe with bootstrap 95% CIs.

Sentiment as fourth expert

FinBERT over NSE filings, gated through the same regime mechanism. Phase 2 deferred plan.

Conclusion

Three takeaways from Phase 3

Foundation models worked where bespoke architectures didn't

- Chronos-LoRA delivered the only positive single-signal Sharpe
- Bespoke architectures failed to generalize out-of-sample
- Pre-trained temporal representations captured regime shifts

Hybridization is conditional, not always-on

- HMM gating helps in stressed regimes
- Gating hurts in clean bull markets
- Ablation makes the trade-off explicit

The engineering ships with the research

- Pinned deps, Docker, manifest md5s, CI
- Phase-grouped dashboard for traceability
- Reviewer can reproduce in 3 commands

Questions?

Thank you.

regime-adaptive-transformer
github.com/sshivanshg